2025 CSC493 – Capstone Weekly Reports[1]

<Download this template and name it Week_<WEEK_NUMBER>, where "<REPORT_NUMBER>" is replaced by the current week number (i.e., "1" for the first weekly report and so forth).

Week	Week 7
Project Name	AI Music Mood Classifier
Report Date	Oct 7, 2025 ← click to select date

Part 1: Weekly Progress Report

- **Accomplishments**: **What did you complete this week?** Be specific and measurable.
- Examples: *Implemented login authentication using OAuth2; Fixed bug in file upload function (issues #14); Completed draft of UI wireframes; Identified functional and non-functional requirements; etc. (up to 3 points)*

This week, I have accomplished creating a rudimentary GUI that allows users to upload music files, load an AI model to classify the song, and output a graph showing the valence and arousal values that the classifier model gathered from the song.

- **Challenges**: **What problems did you encounter?** Mention blockers, unresolved issues, or risks.
- Examples: *Struggled with connecting database migrations to production server; API documentation was incomplete – need clarification; etc. (up to 3 points)*

I encountered a few challenges, mostly dealing with how I should grab the model that the GUI is going to use in a way to where it is pretrained and keeps the data from training for more accuracy, I decided pickling the AI model within the main.py file and grabbing that pickled file in the gui file would be simple and easy to change if issues arise later. I also encountered an issue with the scaler that the baseline classifiers use, as it was adding empty features when applied to the DEAM dataset, but not to user-uploaded music files. I found the issue that caused the empty features to be present and properly set the number of extracted features to 44 for each music file where the features are extracted.

- **Next Steps**: **What are your goals for next week?** Keep them realistic and linked to project milestones.
- Examples: Set up automated testing with GitHub Actions; Final project proposal *(up to 2 points)*

My next steps are to finalize the GUI and enhance its appearance, display more accurate arousal and valence scores (currently rounded to either 0 or 1), and then begin creating a convolutional neural network AI model that will hopefully be more accurate than my current models.

- **Discussion Points**: **What would you like to discuss in class, or with your professor/TA?** This could include technical issues, project direction, or collaboration concerns.
- Examples: *Should we prioritize front-end features before database scaling? Need guidance on writing integration tests; etc. (up to 2 points)*

[1] Detailed Weekly Report requirements can be found here: 2024 URCPP Capstone - Using Agile and Reporting Out

For the GUI, which visualization approach would best communicate mood predictions to end users? Should I aim for a live-plot style (using Matplotlib within Tkinter) or export graphs separately?

Part 2: Time Reporting

Make sure that as you fill out the first prompt, you include in enough detail in the summary. For example, "debugging" is vague, but "debugged function X to make sure that when user does action Y, it is called and returns the value Z" is better.

- **Time Spent**: Briefly explain how much time, *outside of class*, spent on your project. If you worked on multiple components, each should get a detailed summary. Make sure to add up all the hours and minutes correctly. *Add as many rows as you need to the table below. Please do not include hours in class as part of this section. (up to 4 points)*

START	FINISH	HOURS	DETAILED SUMMARY
MM/DD HH:MM	MM/DD HH:MM		
9/30 08:00	9/30 12:00	4	I spent 4 hours fixing issues dealing with the scaler from the model having the incorrect number of extracted feature values that are used to classify the songs. I didn't fully fix the issue on Tuesday.
10/1 15:00	10/1 19:00	4	Fully fixed the wrong number of extracted features issue, and retrained the different models to properly pretrain them, and GUI model call was working.
10/2 10:00	10/2 16:00	6	Worked on the model call and how the arousal and valence scores should be displayed, I choose to have tkinter display a matplotlib graph with four quadrants showing high and low valence and arousal values for the x and y axises respectively.
	WEEKLY TOTAL	14	

- **Total (Cumulative) Project Time Spent**: After the number of hours and minutes, make sure to briefly explain whether you are on track and if not, what you may need to do in order to achieve what you set out to accomplish. *(up to 2 points)*

So far, I have spent 94 hours total working on this project. I am on track according to the timeline built using the help of ChatGPT. During the 7th week of the timeline (October 6 - 13), I need to finalize the GUI, enhancing its appearance, and displaying more accurate arousal and valence scores (not just rounded to either 0 or 1).

Rubric:

The following rubric will be used, but they might change as needed.

Accomplishments (3 points)

1 point for a general description of progress, 2 points for specifics on progress, 3 points for specifics AND referring to previous targets and explaining how current accomplishments build on previous ones.

Challenges (3 points)

1 point for mentioning there are roadblocks, 2 points for specifics, 3 points for specifics AND what was done already to try to overcome them.

Desired discussion points (2 points)

1 point for at least one relevant discussion point as a general question, 2 points for relevant discussion points with specifics

Future Goals (2 points)

1 point for concrete future targets (i.e. "working more on the project" is a zero, but "working on getting component X to interface with component Y" suffices), 2 points for tying in the targets with what was hopefully discussed in the meeting.

Time Spent (4 points)

1 point for including general statements of how much time was spent ("4 hours on coding"), 2 points for splitting time into specific parts ("1.5 hours on research on component X, 1 hour coding, 2.5 hours debugging"), 3 points for specific parts and details on the pieces ("1.5 hours researching Turtle interface for drawing concentric circles given inputs from the user, 1 hour coding function X that used that interface, 2.5 hours testing function X by giving it multiple values and fixing errors for values A, B, C, and D"). 1 Point for totalling the hours correctly.

What happens if your time on a task is interrupted and you don't have a concrete (or discrete) end time? In this case put the start time in, and the word "interrupted" for the end time and include the task total time. Rounding to the nearest 15 minutes is acceptable. *(This makes adding up times easier, especially when you use decimal hours, i.e. 3.75 rather than 3 hours 45 minutes.)*

Total (Cumulative) Project Time (2 points)

1 point for summing the values correctly, 2 points for the total time AND reflection on progress (you are confident to fit the target and if not, what course corrections you anticipate needing to make)

Resources

Here's a link to this Weekly Report Template – Make a copy and use it:

2025 CSC493 Weekly Report v2 TEMPLATE